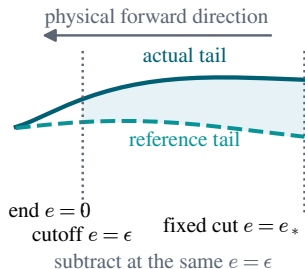


(a) algebraic end: fixed-cut reference



(b) pole end: fixed remaining time

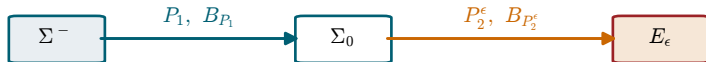
physical remaining time $s = t_b - t$

$$\lambda(X) = \sum_{q=-m_p}^{-1} s^q A_q(\log s, p) + R(s, p)$$



subtract the Laurent--log counterterm $C_{B,p}(\epsilon, p)$

(c) exact action composition is retained by either finite part



finite cutoff: $B_{P_2^\epsilon P_1} = B_{P_1} + B_{P_2^\epsilon} \circ P_1$

subtract $C_{B,e}(\epsilon)$ on the terminal segment, then let $\epsilon \downarrow 0$